H. Maximal AND

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1300 | Tags: bitmasks, greedy, math | https://codeforces.com/problemset/problem/1669/H

Let mathsf*AND* denote the bitwise AND operation, and mathsf*OR* denote the bitwise OR operation.

You are given an array a of length n and a non-negative integer k. You can perform **at most** k operations on the array of the following type:

- Select an index i ($1 \le i \le n$) and replace a_i with a_i mathsf*OR* 2^j where j is any integer between 0 and 30 **inclusive**. In other words, in an operation you can choose an index i ($1 \le i \le n$) and set the j-th bit of a_i to 1 ($0 \le j \le 30$).

Output the maximum possible value of a_1 mathsf*AND* a_2 mathsf*AND* … mathsf*AND* a_n after performing **at most** k operations.

Input

The first line of the input contains a single integer t ($1 \le t \le 100$) — the number of test cases. The description of test cases follows.

The first line of each test case contains the integers n and k ($1 \le n \le 2 \cdot 10^5$, $0 \le k \le 10^9$).

Then a single line follows, containing n integers describing the arrays a ($0 \le a_i$ lt 2^{31}).

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output a single line containing the maximum possible mathsf*AND* value of a_1 mathsf*AND* a_2 mathsf*AND* … mathsf*AND* a_n after performing **at most** k operations.

Example

Input

```
4
3 2
2 1 1
7 0
4 6 6 28 6 6 12
1 30
0
4 4
3 1 3 1
```

Output

```
2
4
2147483646
1073741825
```

Note

For the first test case, we can set the bit 1 (2^1) of the last 2 elements using the 2 operations, thus obtaining the array [2, 3, 3], which has mathsf*AND* value equal to 2.

For the second test case, we can't perform any operations so the answer is just the mathsf*AND* of the whole array which is 4.